Noita-Clone：面向 LLM Agent 认知能力画像的评测环境

A Noita-like Environment for Profiling LLM-Agent Cognition

> **状态：初步设计。**本文内容均为早期想法，后续会继续讨论和修改。
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> **Status: preliminary design.** Everything here is an early idea and will be revised after further discussion.

1. 设计目的 / Purpose

中文：构建一个 Noita 风格的 falling-sand 物理沙盒，以及一套可组合的 spell/wand 系统，作为 **LLM Agent** 的评测环境。评测目标是**能力画像**（agent 在各项认知能力上强弱如何），不是能力上限（通关率）。环境不面向人类玩家，也不涉及 RL 训练。

English: Build a Noita-style falling-sand physics sandbox with a composable spell/wand system as an evaluation environment for **LLM agents**. The goal is a **capability profile** (where an agent is strong or weak across cognitive skills), not a capability ceiling (win rate). The environment is not meant for human players and does not involve RL training.

> **潜在的核心设计：跨局学习。**死亡后世界重置，只有 agent 的记忆保留，借此观察 agent 如何自主做 memory management（记什么、怎么组织、何时检索、如何修正错误记忆）。这一点**尚未敲定**：我们并不是专门测 memory 的 benchmark。不过 Noita 本身非常适合测试 memory，这一点值得记下，留待后续决定其权重。
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> **Potential core design: cross-run learning.** The world resets on death and only the agent's memory persists, letting us observe how agents manage memory on their own (what to record, how to organize it, when to retrieve it, how to correct wrong memories). This is **not yet settled**: we are not building a memory-specific benchmark. Still, Noita is exceptionally well suited to testing memory, which is worth noting for later decisions on its weight.

2. 目标能力 / Target Capabilities

#	能力 / Capability	中文说明	English description
1	**因果推断 / 类比迁移 / 世界模型学习** Causal inference / analogical transfer / world-model learning	三者合并为一类：从观察中归纳规律并迁移到新对象。看到怪物掉进酸里死了，推断自己跳下去也会死。水能灭火，血液、黏液性质相近，推测大部分类水物质都能灭火。	Merged into one: induce regularities from observation and transfer them to new objects. A monster dies in acid, so jumping in would kill me. Water, blood, and slime all extinguish fire, so liquids with water-like properties probably do too.

2	**长期记忆与记忆管理** Long-term memory & memory management	记住魔药配方、wand 配置等并复用。重点观察 agent 自己如何写入、组织、检索和更新记忆。	Retain potion recipes, wand builds, etc. and reuse them. The focus is how the agent itself writes, organizes, retrieves, and updates memory.
3	**长程规划** Long-horizon planning	基于记忆设定并推进长期目标，例如"我的 wand 只缺一个关键 spell，去哪里找"。	Set and pursue long-term goals from memory, e.g. "my wand lacks one key spell, so where do I find it?"
4	**朴素物理直觉** Raw physical intuition	无先验经验时解决新问题，例如第一次意识到可以用水灭火。**必须与长期记忆区分**：前者是第一次想到，后者是已经知道。	Solving novel problems without prior experience, e.g. first realizing water extinguishes fire. **Must be distinguished from long-term memory**: this is first-time insight, not recall.
5	**空间推理** Spatial reasoning	例如理解投射物沿直线飞行，所以只要能看到敌人，理论上就能射到。	E.g. a projectile travels in a straight line, so if I can see the enemy I can in principle hit it.
6	**主动实验与探索** Active experimentation & exploration	对没见过的东西射一发、踢一脚，用行动换取信息。这是**获取**知识的手段，多数其他能力是**使用**知识。	Shoot or kick something unfamiliar to gain information through action. This is how knowledge is **acquired**, while most other capabilities **use** it.
7	**主动感知** Active perception	不是被动接收画面，而是主动研究环境：把光标放到特殊位置获取信息，再据此决策。	Not passive intake: actively probe the environment, e.g. hover the cursor on points of interest to get info, then decide.
8	**组合式程序构造** Compositional program construction	Wand 构建本质上是在搜索一个小型程序（spell 顺序、modifier、trigger、mana 与 cast delay 约束）。	Building a wand is searching over a small program (spell order, modifiers, triggers, mana and cast-delay constraints).
9	**风险决策** Risk-aware decision making	喝未知液体、进未知区域，需要权衡收益与死亡概率。	Drinking unknown liquids or entering unknown areas requires trading off payoff against death risk.
10	**元认知** Metacognition	知道自己不知道什么，所以在稳健的前提下探索。	Knowing what you don't know, and exploring robustly because of it.
11	**链式反应推导** Chain-reaction reasoning	爆炸引燃油，油引爆炸药桶，炸药桶使地形塌陷，agent 掉进火里。要求多步前向推演。	An explosion ignites oil, the oil detonates a barrel, the barrel collapses terrain, and the agent falls into fire. Requires multi-step forward simulation.

12	**反应能力** Reactive ability	躲避弹幕、把握跳跃时机。**待定**：实时、暂停等待，还是固定 tick？对 LLM agent 而言，这决定该维度测的是反射，还是推理延迟下的表现。	Dodging projectiles, timing jumps. **Undecided**: real-time, pause-and-wait, or fixed-tick? For LLM agents this decides whether the dimension measures reflexes or performance under reasoning latency.
13	**组合能力** Composite capabilities	高阶行为由多种能力组合而成，例如资源规划 = 长期记忆 + 长程规划 + 少量推理。	High-level behavior is a composition of skills, e.g. resource management = long-term memory + long-horizon planning + some reasoning.

能力的网络化结构 / Capabilities as a network

中文：能力不是平铺的清单，而是一张图。基础能力（如感知、因果推断、空间推理、记忆）是节点，组合能力是依赖多个节点的上层结构。这张图后续可以继续细化：把粗粒度能力拆成更细的子能力，并让任务标注自己调用了哪些节点。这样既能做单点诊断，也能解释"组合任务失败，是哪个底层节点出了问题"。

English: Capabilities form a graph, not a flat list. Basic capabilities (perception, causal inference, spatial reasoning, memory) are nodes, and composite capabilities are higher-level structures depending on several nodes. The graph can be refined further by splitting coarse capabilities into finer sub-capabilities, with each task annotated by the nodes it exercises. This supports both single-node diagnosis and explaining which underlying node caused a composite task to fail.

3. Noita 为什么难 / Why Noita Is Hard

中文	English
知识惩罚：不知道某个机制或危险，就会死。	**Knowledge penalty**: not knowing a mechanic or hazard gets you killed.
局外成长只在脑子里：死亡后唯一保留的是知识。	**Meta-progression lives only in your head**: knowledge is the only thing that persists across deaths.
弱引导：几乎没有目标指引或教程。	**Minimal guidance**: almost no objectives or tutorials.
高不确定性与动态环境：程序生成加完整物理模拟，没有两个情境完全相同。	**High uncertainty, dynamic world**: procedural generation plus full physics simulation means no two situations repeat.
不可逆且永久死亡。	**Irreversible actions and permadeath**.
稀疏且延迟的反馈：大部分行为没有即时对错信号。	**Sparse and delayed feedback**: most actions give no immediate right-or-wrong signal.
组合爆炸：spell 组合、材料组合都是指数级。	**Combinatorial explosion**: spell and material combinations grow exponentially.

4. Benchmark 设计原则（初步） / Design Principles (Preliminary)

> **重点：以下原则仅为初步想法，尚未定稿，之后会再细聊。**
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> **Key point: the principles below are early ideas, not finalized, and will be discussed further.**

中文	English
防数据污染（扰动）：Noita 的 wiki 很可能在训练数据里。用 seed 随机配方，并对材料属性、spell 名称和效果做受控扰动，使记忆原作知识无益甚至有害。	**Contamination control (perturbation)**: the Noita wiki is likely in training data. Use seeded recipes and controlled perturbation of material properties and spell names/effects, so memorized original-game knowledge gives no advantage, or even misleads.
与原作及同类 Roguelike 拉开距离：玩法与规则尽量不同于 Noita 和典型 Roguelike，避免 agent 靠类型套路取胜；同时用消融实验验证成绩来自目标能力，而非套路。	**Distance from the original and typical roguelikes**: keep mechanics and rules distinct from Noita and generic roguelikes so agents can't win on genre templates, and use ablations to check that scores come from the target capabilities.
能力画像优先：能力隔离的微任务为主，完整对局为辅；每个任务标注调用了能力图中的哪些节点。	**Profile first**: capability-isolating micro-tasks are primary, full runs secondary; each task is annotated with the capability-graph nodes it exercises.
记忆形态要区分：参数记忆（预训练）、上下文记忆（同一局内）、外部记忆（跨局笔记）测的是不同的东西，要分别可测。	**Distinguish memory types**: parametric (pretraining), in-context (within a run), and external (cross-run notes) measure different things and must be separately measurable.
可开关的消融维度：是否可访问 wiki、是否有跨局记忆、是否可暂停、观察是像素还是结构化数据。	**Switchable ablations**: wiki access, cross-run memory, pause support, pixel vs structured observation.
引擎支撑：确定性 seed、headless 模式、存档与回滚（用于带 ground truth 的反事实测试）、事件日志、可变速、面向 agent 的观察与动作 API。	**Engine support**: deterministic seeds, headless mode, save/restore (for counterfactual tests with ground truth), event logs, adjustable speed, an agent-facing observation/action API.
仿真内核与前端解耦：环境不为人类游玩而设计，渲染只服务于调试与可视化。	**Sim core decoupled from frontend**: the environment is not built for human play, and rendering serves only debugging and visualization.